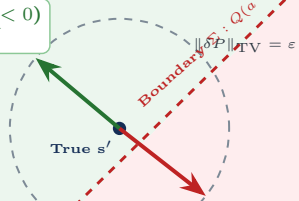


### (a) Directional Geometry & Mismatch

**Region  $\mathcal{R}(a^*)$**   
 $\pi^*(s) = a^*$  (Optimal)

**Expansive ( $B < 0$ )**  
 Safe ( $C_i = 0$ ) ✓

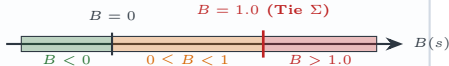


**Compressive ( $B > 1$ )**  
 Action Flip ( $Z = 1$ ) ✗

**Region  $\mathcal{R}(a^{(2)})$**   
 $\pi(s) = a^{(2)}$  (Suboptimal)

**Objective Mismatch:** Identical  $\|\delta P\|_{TV}$  error yields opposite outcomes ( $C_i = 0$  safe vs.  $Z = 1$  flip).

### (b) Margin Gauge & Four Regimes



**1. Expansive ( $B < 0$ )** [✓]  
 $\Delta m > 0 \Rightarrow m_{\hat{P}} > m_P$   
 Optimal  $a^*$  reinforced.

**2. Sub-thresh ( $0 \leq B < 1$ )** [≈]  
 $-m_P < \Delta m \leq 0 \Rightarrow m_{\hat{P}} > 0$   
 Optimal  $a^*$  preserved.

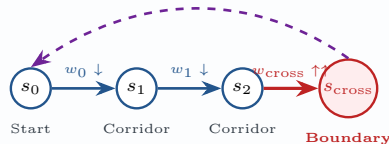
**4. Crossing ( $B > 1.0$ )** [✗]  
 $\Delta m < -m_P \Rightarrow \hat{Q}(a^{(2)}) > \hat{Q}(a^*)$   
**Action Flip:**  $Z_{\text{cross}} = 1$ .

**3. Tie ( $\Sigma$ ) ( $B = 1.0$ )** [Tie]  
 $\Delta m = -m_P \Rightarrow m_{\hat{P}} = 0$   
 Degenerate:  $\hat{Q}(a^*) = \hat{Q}(a^{(2)})$ .

**Diagnostic Metric:**  $B(s) = -\Delta m(s)/m_P(s)$  is the sharp causal parameter for action failure.

### (c) Spatial Corridor & Propagation

**Global Backup:**  $V(s) \leftarrow \max_a [r + \gamma \mathbb{E}V(s')]$



**1. Boundary Up-weight**  
 $w(s_{\text{cross}}) \uparrow (\lambda > 0)$

**2. Corridor Depletion**  
 $w(s_1, s_2) \downarrow (\sum w_i = N)$

**4. Upstream Policy Flips**  
 Action flips at  $s_0 \downarrow$

**3. Value Distortion**  
 $\hat{V}(s')$  backup noise

**Bottleneck:** Sample reallocation starves corridor dynamics, propagating errors to invert upstream actions.